

Quantum Kernel Alignment for Clinical Data Classification

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Introduction

In recent years, different quantum machine learning algorithms have been developed that could lead to quantum speed-ups over classical machine learning approaches. Apart from a proposed potential speed-up, it remains unclear, whether the quantum algorithmic properties could leverage other challenges. Sparse heterogeneous datasets are a relevant challenge in many fields, leading to these datasets to be excluded from reliable computing. Here, utilizing the expressivity of the quantum (-inspired) kernel, we construct a classification workflow that is allowing for the first time to robustly deal with heterogeneous limited noisy datasets in a clinical setting and compare it to a classical Radial Basis Function (RBF)-kernel Support Vector Machine (SVM).

Quantum Kernel Methods

In the case of QKMs, the core component that can result in outperforming classical learners is the *quantum feature map* $\Psi(x) = |\psi(x)\rangle \langle \psi(x)|$ with $|\psi(x)\rangle = U(x) |0^{\otimes n}\rangle$. The quantum kernel entry between two data points x_i and x_j can be estimated by preparing $U(x_i) |0^{\otimes n}\rangle$, applying $(U^\dagger(x_j))$, and measuring $|\langle 0^{\otimes n} | U^\dagger(x_j) U(x_i) | 0^{\otimes n} \rangle|^2$. This precomputed kernel can then be passed to an SVM. The used feature map is

$$|\phi_\theta(x_i)\rangle = \left(\prod_{j=1}^n e^{-i \frac{t}{T} x_{ij} H_j^{XYZ}} \right)^T U_{\text{ent}}^L \bigotimes_{j=1}^{n+1} |0^{\otimes n}\rangle \quad (1)$$

with the trainable parameters $\theta = (T, t, L)$, $H_j^{XYZ} = (X_j X_{j+1} + Y_j Y_{j+1} + Z_j Z_{j+1})$, the Pauli operators X_j, Y_j, Z_j acting on the j -th qubit and L the number of strongly entangling layers.

Clinical Data Analysis

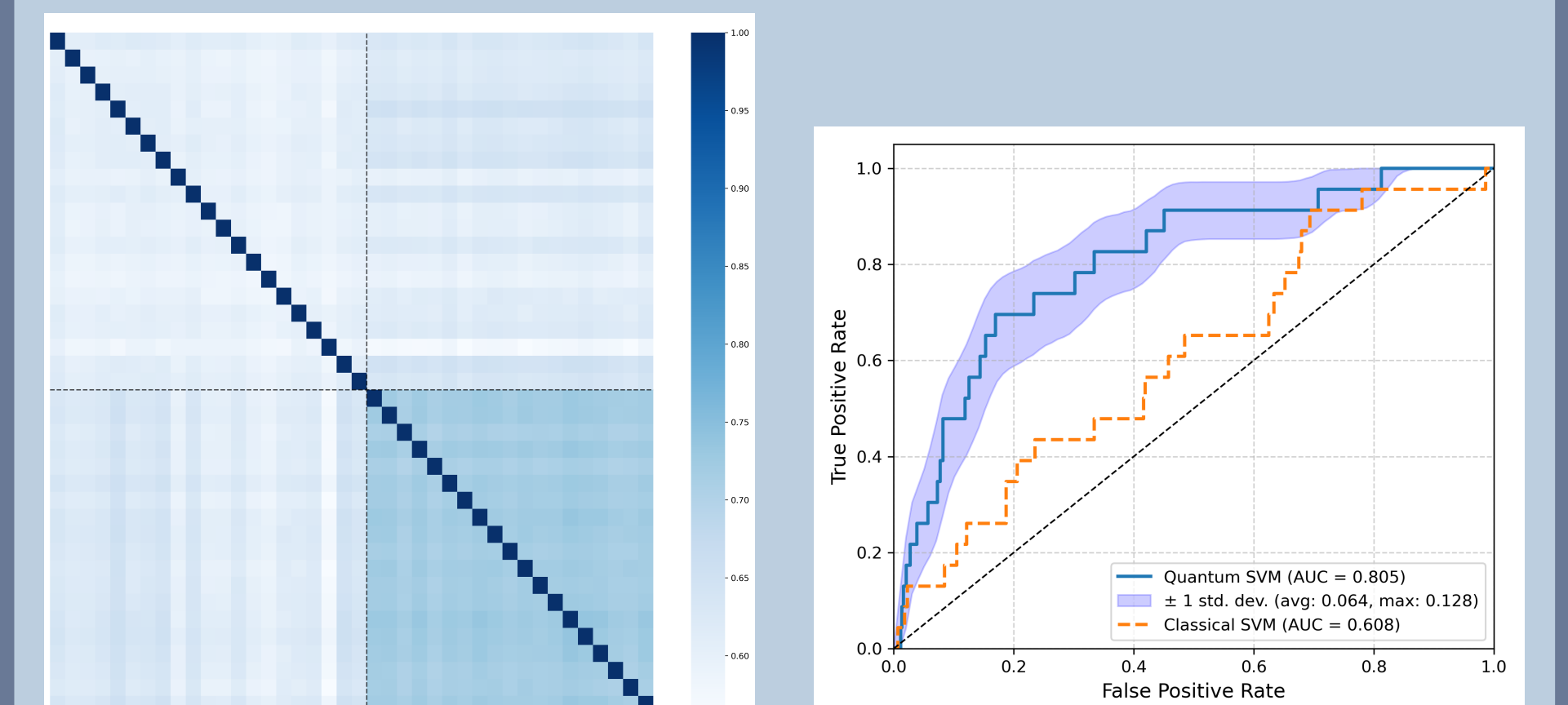


Fig 2: Kernel matrix (left) and ROC curve (right) for the stroke dataset.

Datasets and Workflow

Fig 2 Stroke prediction dataset from kaggle.

Fig 3 Pancreatic Ductal Adenocarcinoma (PDAC) is one of the most deadly tumors with practically no existing therapy.

Fig 4 “Cold-MSS” refers to tumors that are microsatellite stable (MSS) and immunologically “cold”, meaning they show little immune activity and typically do not respond to checkpoint-inhibitor therapy.

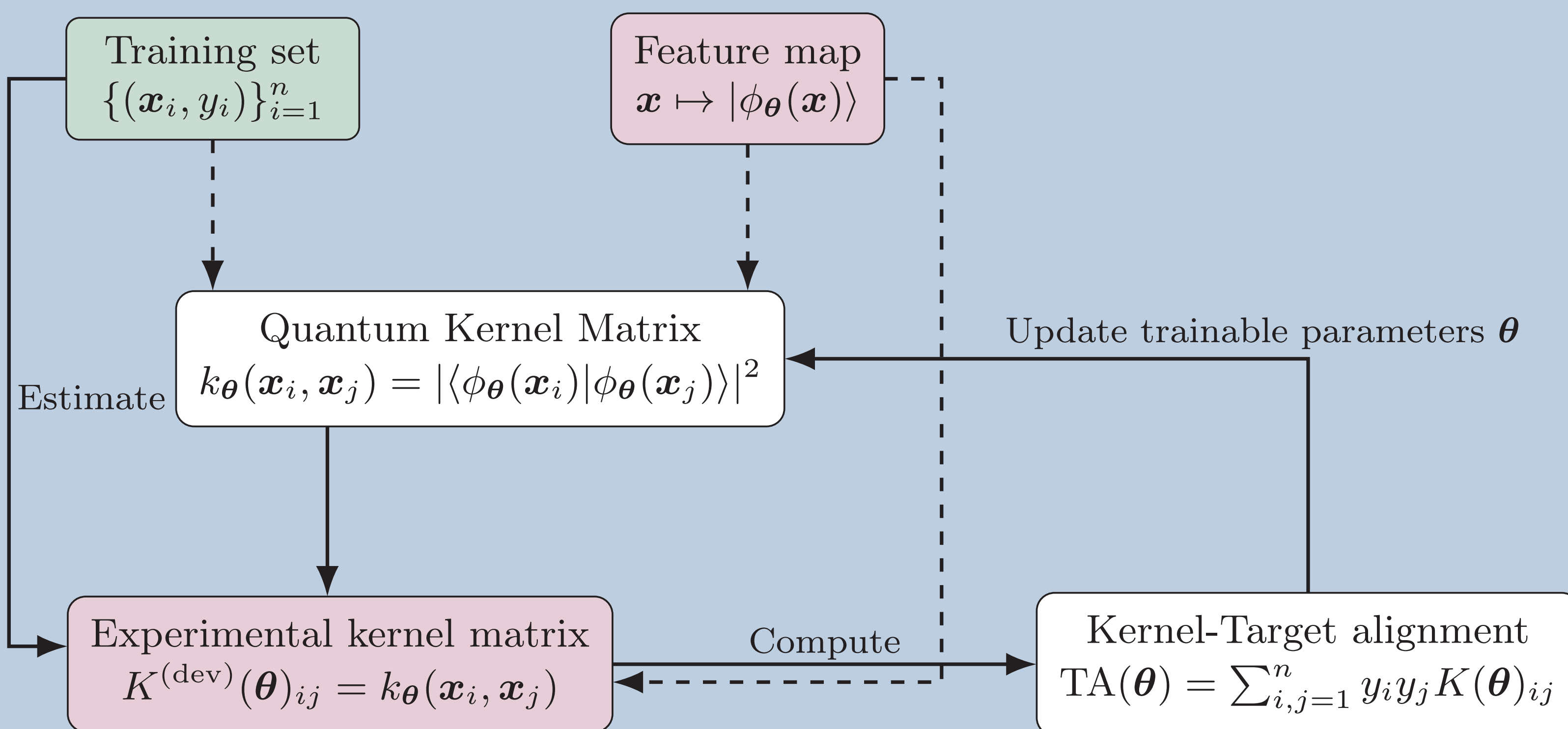


Fig 1: Quantum kernel alignment procedure

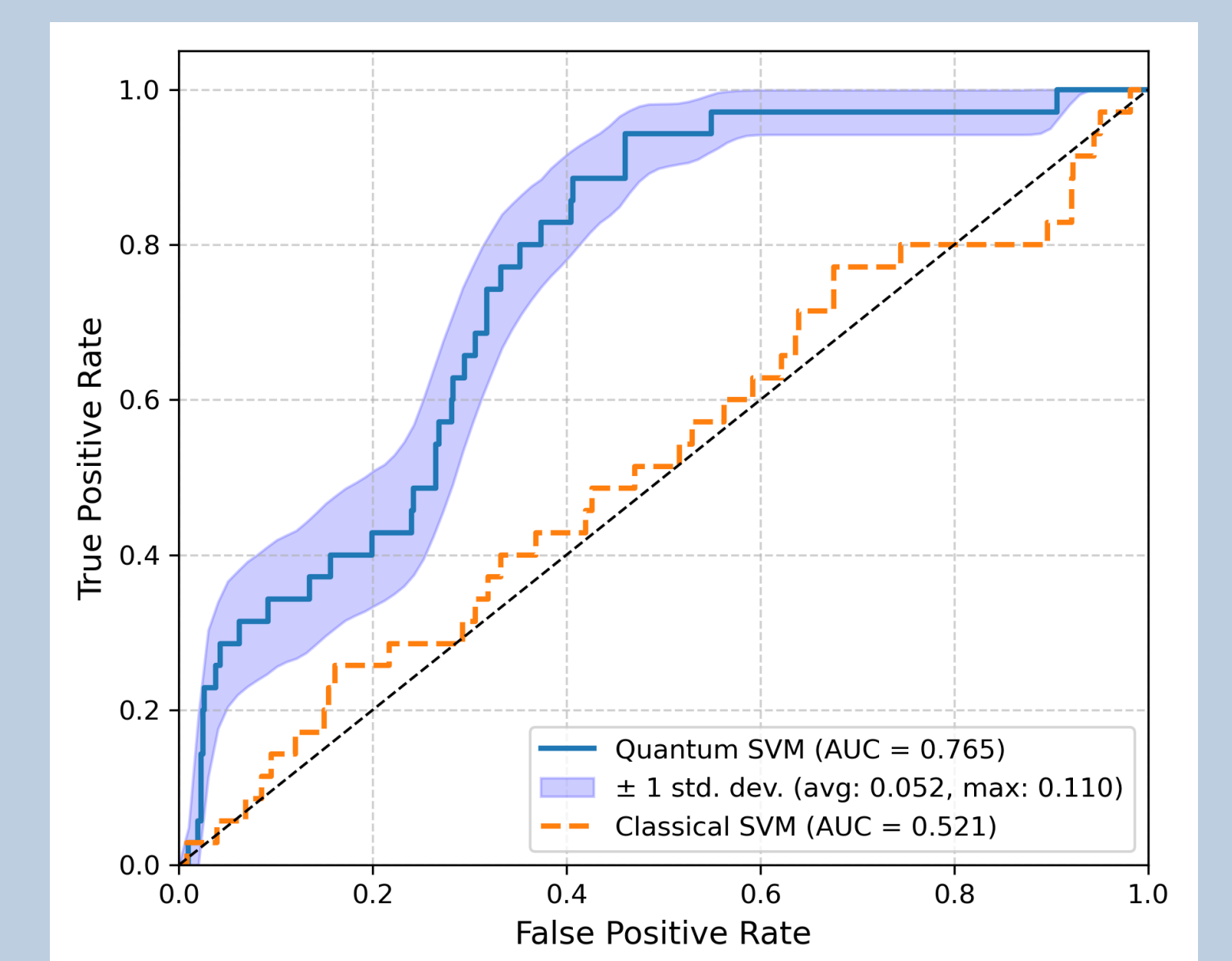


Fig 3: PDAC ROC-Curve

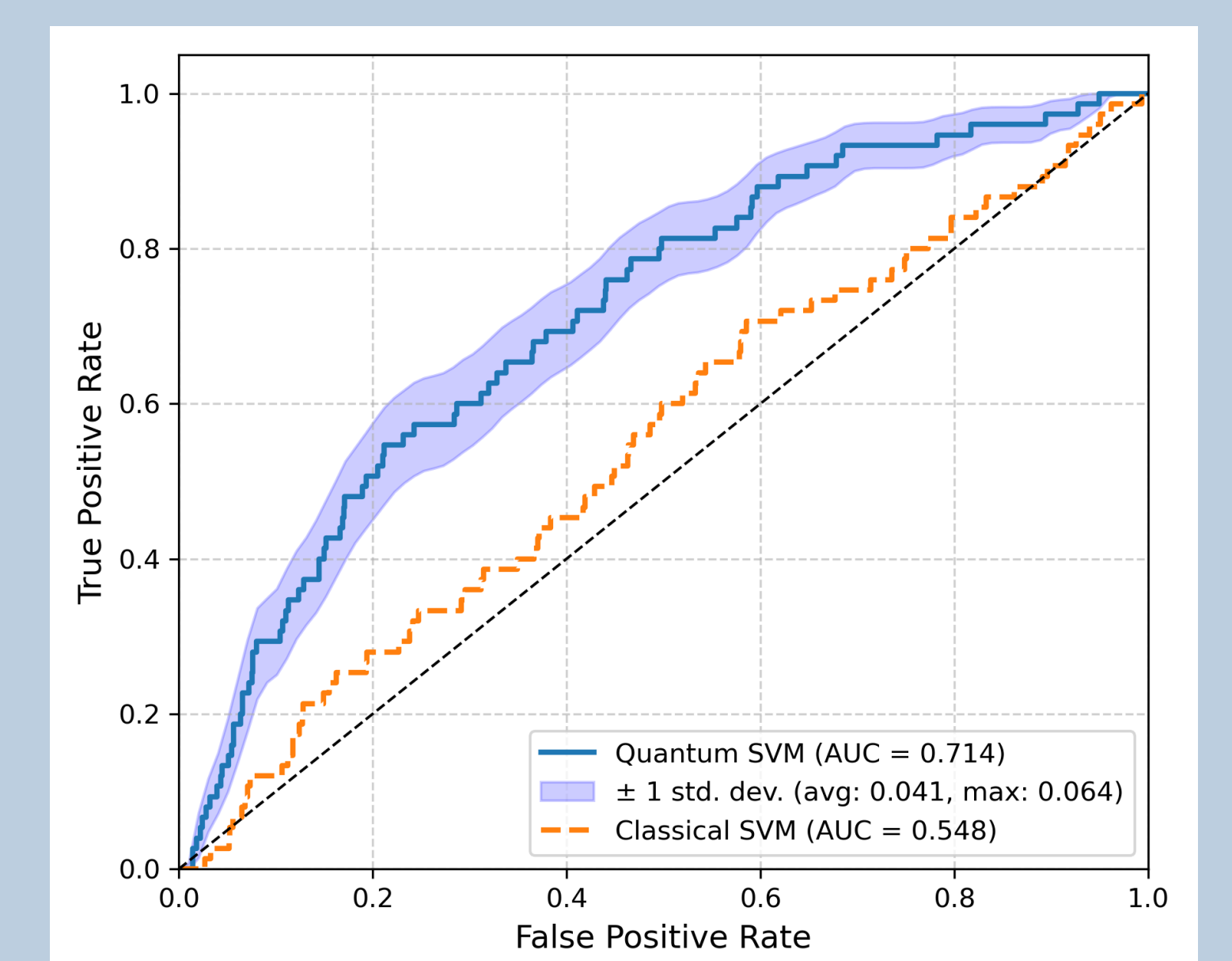


Fig 4: Cold-MSS ROC-Curve

Conclusion

Quantum kernel methods (QKMs) provide an effective approach to analyzing complex clinical datasets, especially when data availability is limited. The results show that the choice of the feature map can significantly affect classification performance, particularly in identifying patterns within the minority class. By leveraging quantum inspired kernels, QSVMs can capture and separate high-dimensional, nonlinear data patterns, with small sample sizes- outperforming classical RBF-SVMs. This makes QKMs a promising tool for clinical applications where obtaining large datasets is challenging and subtle data relationships are directly linked to patient survival.

References

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Affiliations

